

VENTERSDORP, South Africa

WMO: 683490

Lat: 26.312S

Lon: 26.814E

Elev: 1494

StdP: 84.62

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	0.4	1.9	-12.0	1.6	14.3	-9.9	1.9	13.3	18.4	17.1	15.8	16.5	3.1	20	0.444

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.5	32.1	16.1	30.8	15.8	29.5	15.8	19.7	25.5	19.3	25.0	18.8	24.5	5.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.2	15.8	21.0	17.8	15.4	20.8	17.2	14.7	20.5	63.8	25.5	61.9	24.9	60.2	24.7	21.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.2	13.4	11.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.6	21.9	21.7	19.9	17.3	14.0	11.4	11.3	13.9	17.9	20.0	20.4	21.8
	DBStd	4.73	2.57	2.16	2.10	2.61	2.80	2.35	2.63	3.11	3.47	3.36	3.17	2.53
	HDD10.0	43	0	0	0	1	7	11	17	7	1	1	0	0
	HDD18.3	852	4	3	8	48	136	207	217	139	48	22	15	4
	CDD10.0	2816	369	327	308	218	129	54	57	128	237	311	312	364
	CDD18.3	584	115	96	58	17	1	0	0	2	34	74	77	109
	CDH23.3	4717	776	634	393	137	32	1	1	63	408	719	731	823
CDH26.7	1326	237	183	68	17	1	0	0	3	102	230	230	256	
Wind	WSAvg	4.6	4.6	4.8	3.4	3.7	3.8	4.0	4.2	5.2	5.3	5.4	5.3	5.1
Precipitation	PrecAvg	602	102	80	82	42	20	6	1	5	14	56	66	101
	PrecMax	873	186	188	214	128	98	41	7	24	75	190	182	154
	PrecMin	391	28	22	17	1	0	0	0	0	0	7	8	35
	PrecStd	122	45	42	44	30	25	9	2	7	21	42	37	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.0	32.6	30.1	28.6	25.7	22.8	23.2	27.0	30.6	32.7	33.7	33.7
		MCWB	17.0	16.9	15.8	15.6	11.8	10.0	10.4	13.0	13.7	14.6	14.8	16.9
	2%	DB	30.9	30.7	28.5	26.5	24.1	20.9	21.2	25.1	29.1	31.1	31.1	31.3
		MCWB	17.8	16.9	16.2	15.1	12.1	9.7	9.8	11.8	13.2	14.3	14.8	16.8
	5%	DB	29.1	28.9	27.1	24.9	22.8	19.2	19.8	23.1	27.7	29.4	29.3	29.3
		MCWB	17.8	17.4	16.5	14.6	11.8	9.2	9.3	10.6	12.7	14.2	14.9	16.9
	10%	DB	27.4	27.2	25.7	23.3	21.1	17.7	18.1	21.1	25.8	27.6	27.7	27.7
		MCWB	17.9	17.6	16.4	14.2	11.1	8.8	8.9	9.8	12.2	14.1	15.1	16.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.6	20.1	19.3	18.1	14.4	12.0	11.4	14.6	15.5	17.6	19.4	20.4
		MCDB	26.2	26.4	25.1	23.2	22.1	18.7	19.9	23.8	26.0	25.7	24.7	25.9
	2%	WB	19.8	19.4	18.6	17.1	13.4	11.0	10.6	13.0	14.6	16.7	18.2	19.6
		MCDB	25.2	25.7	23.7	22.7	21.2	17.9	19.2	22.4	24.8	24.9	24.8	25.7
	5%	WB	19.3	18.8	18.0	16.4	12.6	10.2	10.0	11.9	13.9	16.0	17.6	19.0
		MCDB	24.6	25.1	22.7	22.0	20.4	17.0	18.5	21.1	24.3	24.7	24.1	25.1
	10%	WB	18.8	18.3	17.4	15.6	11.8	9.4	9.2	10.8	13.2	15.3	16.9	18.2
		MCDB	24.0	24.4	22.2	21.1	19.2	16.0	17.2	19.5	22.9	23.8	23.1	24.4
Mean Daily Temperature Range	5% DB	MDBR	10.5	11.0	11.2	12.6	14.5	14.4	14.6	14.9	14.7	13.4	12.8	11.3
		MCDBR	13.5	13.7	13.6	14.2	16.5	15.9	16.0	16.6	16.6	15.2	14.8	13.8
		MCWBR	4.4	4.9	4.7	5.6	7.1	7.5	7.6	7.5	6.9	5.6	5.3	4.8
	5% WB	MCDBR	10.6	11.4	10.6	12.3	14.3	14.1	15.1	15.5	14.8	13.5	11.9	11.2
		MCWBR	4.3	4.7	4.4	5.7	6.8	7.2	7.7	7.7	6.6	5.6	5.2	4.6
Clear-Sky Solar Irradiance	taub	0.371	0.376	0.361	0.319	0.283	0.267	0.268	0.320	0.362	0.391	0.346	0.355	
	taud	2.374	2.355	2.377	2.466	2.542	2.555	2.529	2.373	2.265	2.223	2.403	2.408	
	Ebn at Noon	972	951	934	932	928	925	937	915	918	925	990	989	
	Edh at Noon	131	130	121	100	84	79	84	108	133	147	126	127	
All-Sky Solar Radiation	RadAvg	6.80	6.32	5.63	4.79	4.35	3.93	4.28	5.09	6.14	6.63	6.99	7.08	
	RadStd	0.71	0.58	0.49	0.41	0.20	0.18	0.12	0.25	0.25	0.43	0.57	0.42	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days		
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	+0.66	+0.75	N/A	N/A	N/A	N/A	N/A	N/A	-181	+234

Nomenclature: See separate page